# Smart Women Safety Wearable Using ESP32-CAM With Real-Time Evidence Capture and GSM-Based Emergency Alerts

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***Abstract*—This research presents a wearable personal safety system designed using an event-driven emergency detection architecture to enhance rapid response during critical situations. Increasing incidents of harassment, assault, and personal security threats highlight the need for autonomous safety mechanisms that do not rely solely on manual user interaction. Conventional smartphone-based safety applications often require multiple steps for activation and may fail when the victim is unable to access or operate the device [3]. To address these limitations, the proposed system introduces a compact wearable solution capable of automatically detecting potential emergency scenarios and transmitting real-time evidence to trusted contacts.**

**The system implements three distinct trigger mechanisms: manual SOS button activation, loud-sound distress detection using a microphone sensor, and tamper or forced removal detection using a vibration sensor [5], [6]. These triggers are integrated within an event-driven architecture that enables the system to transition from a monitoring state to an emergency response state whenever a valid event is detected. Upon activation, the ESP32-CAM module [1] captures real-time images of the surrounding environment and embeds contextual metadata including GPS coordinates obtained from a NEO-6M module and accurate timestamps from a DS3231 real-time clock. The collected data is then packaged and transmitted instantly through the Telegram Bot API [2] to a pre-configured emergency contact, ensuring rapid communication and secure remote storage of evidence.**

**Experimental evaluation of the prototype demonstrates reliable operation under multiple emergency scenarios, with an average alert transmission delay of only a few seconds depending on network conditions. The proposed system provides multi-modal evidence generation, tamper-resilient alert transmission, and autonomous detection capabilities while maintaining a low-cost and portable design [7], [12]. Due to its scalability and ease of deployment, the system can be applied in various contexts including women's safety, elderly monitoring, lone worker protection, and emergency incident documentation.**

***Index Terms*—Wearable Safety Device, IoT, ESP32-CAM, Tamper Detection, Loud-Sound Detection, Telegram API, GPS Evidence Transmission, Event-Driven Architecture.**

## I. Introduction

Personal security threats such as harassment, kidnapping, and assault continue to be reported across both urban and semi-urban environments. In many real-world situations, victims may not have sufficient time or ability to seek help through conventional means. Modern smartphone-based safety mechanisms, such as panic applications or emergency contact dialers, require manual interaction for activation [3]. Under high-stress conditions, however, victims may be physically restrained, unconscious, or unable to unlock and operate their smartphones. As a result, these systems may fail to provide timely assistance during critical moments.

In recent years, increasing awareness of women's safety and personal security has led to the development of several digital safety platforms [8], [11], [12]. Although these applications allow users to share their location or send emergency notifications, they still rely heavily on active user participation. Furthermore, such solutions often provide limited contextual information about the emergency, typically transmitting only location data without visual evidence or detailed event descriptions. This limitation can delay response time and complicate post-incident investigation.

Rapid advancements in Internet of Things (IoT) technologies have opened new possibilities for designing intelligent personal safety systems [3], [4]. By integrating sensors, communication modules, and embedded computing platforms, wearable devices can continuously monitor environmental conditions and automatically detect distress situations. These devices are capable of transmitting critical information such as geographic location, timestamps, and visual evidence, thereby improving situational awareness and enabling faster emergency response.

In addition, modern urban lifestyles often involve late-night commuting, remote workplaces, and increased individual mobility, which further elevate personal safety risks [9], [15]. These circumstances highlight the growing demand for compact wearable safety technologies that can operate autonomously and provide immediate alerts without requiring complex user interaction.

To address these challenges, this paper proposes a wearable emergency detection system based on an ESP32-CAM module [1] and an event-driven architecture. The system integrates three primary emergency triggers: manual SOS activation, loud-sound distress detection using a microphone sensor, and tamper or forced removal detection using a vibration sensor [5]–[7]. When any of these triggers are activated, the system automatically transitions from a monitoring state to an emergency response state. Upon detection of an emergency event, the ESP32-CAM captures real-time images and collects contextual metadata including GPS coordinates and accurate timestamps [4]. This information is then packaged into a structured alert message and transmitted instantly to a pre-configured emergency contact through the Telegram messaging platform [2].

## II. Evolution of Personal Safety Technologies

Personal safety technologies have evolved significantly over the past decade, driven by advancements in mobile computing, sensor networks, and IoT platforms [12].

Early safety solutions primarily consisted of manual panic buttons or emergency helpline services [17]. These systems required the victim to actively trigger an alert, which limited their effectiveness in situations involving physical restraint or sudden attacks.

The next stage of development introduced smartphone-based safety applications that allowed users to share their location with trusted contacts [18]. Although these applications improved accessibility, they still depended on user interaction and often lacked contextual information such as visual evidence or event classification.

Recent developments in wearable technology have enabled the integration of sensors, wireless communication modules, and embedded cameras into compact devices [7], [13]. These systems can continuously monitor environmental conditions and detect potential emergencies using sensor triggers. However, many existing wearable solutions focus primarily on health monitoring or simple location tracking [6], [9].

The current research trend focuses on autonomous safety systems that combine multiple sensing modalities with intelligent communication frameworks [4], [8], [14]. By incorporating real-time image capture, location tracking, and automated alert transmission, modern IoT-based wearable devices provide richer situational awareness and stronger evidence for emergency response and investigation.

## III. Related Work

Several personal safety mechanisms have been introduced in recent years, including smartphone-based SOS applications, GPS trackers, location-sharing platforms, and surveillance systems [12]. While these approaches provide a certain level of protection, they often exhibit significant limitations when applied to real emergency situations. Smartphone-based safety applications typically require the user to unlock the phone, open the application, and manually trigger an alarm or share their location [17]. In high-stress scenarios where the victim is physically restrained, injured, or under immediate threat, performing such multi-step actions may not be possible.

GPS tracking devices are capable of continuously transmitting location information; however, they generally provide only positional data without any contextual evidence regarding the surrounding environment or the nature of the emergency [15]. Similarly, closed-circuit television (CCTV) systems offer video monitoring but are limited to fixed installations and cannot ensure continuous personal coverage. Panic buttons installed in public transportation systems or buildings may also fail to provide reliable assistance, as they can be disabled, ignored, or inaccessible during critical situations [16], [17].

Furthermore, most commercially available wearable devices are primarily designed for health monitoring functions such as heart-rate tracking or step counting rather than for safety-oriented emergency detection and incident documentation [13]. Due to these limitations, many existing safety solutions lack autonomous emergency detection, contextual evidence generation, and reliable incident classification [8], [11]. Most systems depend heavily on manual user activation, and the alerts they generate typically contain only location information without visual confirmation of the event.

In contrast, the proposed system introduces a wearable safety device that integrates multiple autonomous emergency detection mechanisms along with contextual evidence generation. The system incorporates three independent trigger mechanisms: manual SOS activation, loud-sound distress detection, and tamper or forced removal detection [5]–[7]. These triggers enable the system to detect both user-initiated and automatically inferred emergency situations. Upon activation of any trigger, the ESP32-CAM module [1] captures real-time environmental images and associates them with GPS-based location data and RTC-based timestamps. This information is packaged into a structured alert and transmitted through the Telegram messaging platform [2], ensuring rapid delivery and secure remote storage of evidence. By combining event-driven detection, real-time image capture, location tracking, and tamper monitoring, the proposed system provides multi-modal situational evidence that significantly improves emergency response effectiveness and incident documentation compared with conventional safety solutions.

## IV. System Architecture

The proposed wearable safety device is designed using an event-driven architecture in which system responses are triggered by specific emergency events. Under normal operating conditions, the system continuously monitors sensor inputs while remaining in a low-activity monitoring state. When a predefined trigger event is detected, the system immediately transitions into an emergency response state and initiates the alert generation process.

The ESP32-CAM module [1] serves as the central controller of the system, integrating sensing, processing, image capture, and wireless communication functionalities. Multiple peripheral components are connected to the ESP32-CAM to enable comprehensive monitoring of potential emergency conditions.

### *A. Hardware Components*

The wearable safety device consists of several hardware modules that work together to detect emergencies, collect contextual information, and transmit alerts.

- **ESP32-CAM** [1]: Acts as the primary microcontroller unit. It integrates a microcontroller, camera module, and Wi-Fi communication capability within a compact platform. Responsible for monitoring sensor inputs, capturing real-time images, processing emergency events, and transmitting alert messages to the designated emergency contact.

- **GPS Module (NEO-6M)**: Provides real-time geographical location information in the form of latitude and longitude coordinates. These coordinates are converted into a Google Maps link, allowing emergency contacts to quickly identify the exact location of the wearer [15].
- **RTC Module (DS3231)**: Supplies accurate date and time information for each detected emergency event. The timestamp is embedded within the alert message to ensure precise temporal documentation of the incident.
- **Microphone Sensor**: Detects high-intensity distress sounds such as shouting or calls for help. An alert is triggered when the sound intensity exceeds a predefined threshold value [14].
- **SW-420 Vibration Sensor**: Detects sudden movements or vibrations that may indicate forced removal or tampering of the wearable device [10].
- **SOS Button**: Enables the wearer to manually activate an emergency alert when conscious and able to respond. Pressing the button immediately triggers the emergency response sequence [17].
- **Battery System**: A rechargeable battery ensures portability and continuous operation without requiring constant external power sources.

### B. Event-Driven Logic

The proposed system operates using an event-driven monitoring mechanism in which the ESP32-CAM [1] continuously observes sensor inputs to detect potential emergency conditions. During normal operation, the system remains in a monitoring state where it periodically checks the status of three primary inputs: the SOS push button, the microphone-based sound detection sensor, and the SW-420 vibration sensor [5]. Each detected trigger is mapped to a corresponding emergency classification:

- SOS button activation → *SOS Emergency*
- Loud sound detection → *Loud Sound Distress* [14]
- Vibration detection → *Tamper Alert* [10]

When any of these trigger conditions is detected, the system transitions from the normal monitoring state to an emergency response state. In this state, the ESP32-CAM initiates an automated sequence of operations. First, the camera captures an image of the surrounding environment to provide visual evidence of the situation [4]. Next, the system retrieves the current geographic coordinates from the GPS module and obtains an accurate timestamp from the RTC module. These pieces of information are then combined into a structured alert message that includes the emergency type, date and time of occurrence, and a Google Maps location link. Finally, the alert message along with the captured image is transmitted to a pre-configured Telegram contact or group using the Telegram Bot API [2].

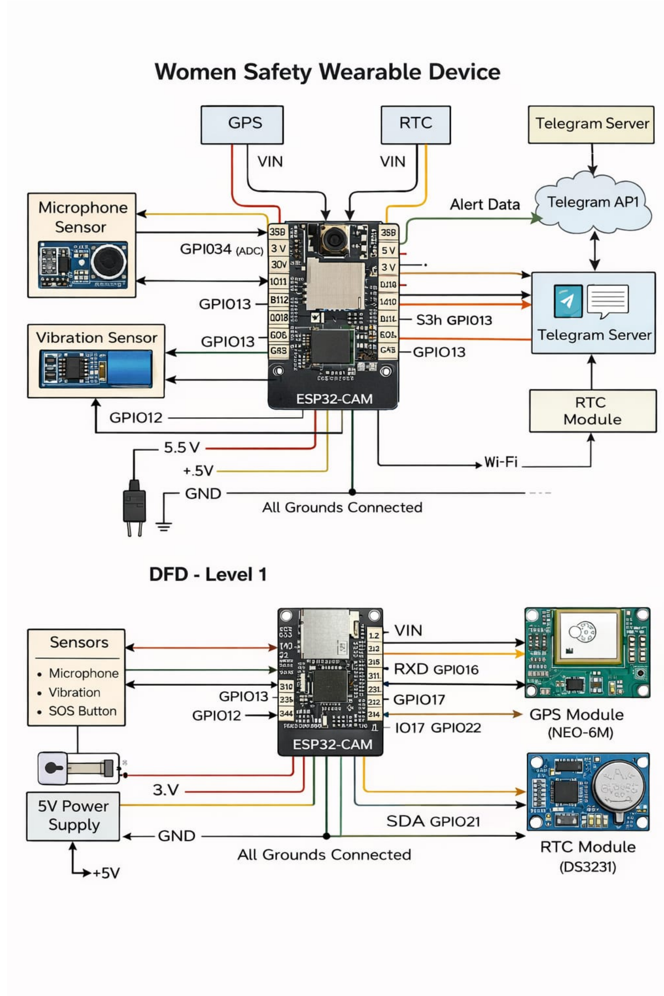


Fig. 1. Proposed System Circuit Diagram

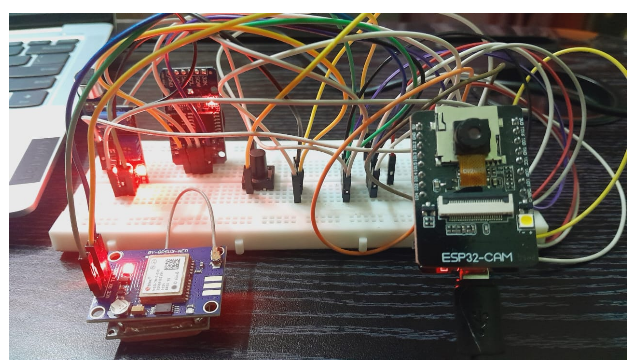


Fig. 2. System architecture of the proposed wearable emergency detection device.

### C. Software and Communication Stack

The software architecture of the proposed system is implemented through embedded firmware running on the ESP32-CAM microcontroller [1]. The firmware coordinates the interaction between sensors, peripheral modules, and communication interfaces to enable reliable emergency detection and alert transmission. The communication framework utilizes the Wi-Fi capability of the ESP32-CAM to transmit alerts through the Telegram Bot API [2]. This approach enables real-time delivery of emergency notifications and allows the captured data to be securely stored on remote cloud servers associated with the Telegram platform. The firmware performs the following sequence of operations:

1) Initialization of system components including the camera module, microphone sensor, vibration sensor, GPS module, and RTC module.
2) Continuous monitoring of event triggers using polling or interrupt-based detection to identify SOS activation, loud-sound detection, or tamper events.
3) Image acquisition using the ESP32-CAM camera module immediately after a valid emergency trigger is detected [4].
4) Retrieval of contextual information including GPS latitude and longitude coordinates and the corresponding timestamp from the RTC module.
5) Generation of a structured alert message containing the emergency classification, date and time of occurrence, and a Google Maps link representing the current location.
6) Transmission of the alert message along with the captured image to a pre-configured Telegram contact or group using HTTP-based requests through the Telegram Bot API [2].

The Telegram communication is implemented using HTTP-based requests to the Telegram Bot API [2]. Once the data is transmitted, the message and media remain stored on the Telegram cloud servers, providing tamper-resilient evidence even if the device is later damaged or destroyed.

## V. Working Principle

When the wearer presses the SOS push button or when the microphone sensor detects a high-intensity distress sound such as shouting or calls for help [14], the system immediately transitions from the monitoring state to the emergency response state. The ESP32-CAM module [1] captures a real-time image of the surrounding environment to provide visual evidence of the situation. Simultaneously, the system retrieves the current geographic coordinates from the GPS module and obtains the exact date and time from the real-time clock (RTC) module. The ESP32-CAM then generates an automated alert message indicating the cause of activation, such as “SOS Emergency Detected” or “Loud Sound Distress Detected”. The complete alert message, together with the captured image and location link, is transmitted to a pre-configured Telegram contact or group [2], ensuring that emergency contacts receive both visual and contextual information within seconds, enabling faster interpretation of the situation and prompt response.

### *A. Tamper Detection*

In scenarios where an attacker attempts to remove, damage, or forcibly detach the wearable device, the SW-420 vibration sensor detects abrupt motion patterns associated with tampering [10]. When such abnormal vibrations are detected, the system activates a tamper-alert mode. The ESP32-CAM captures an image of the surrounding environment and records the last known GPS location along with the corresponding timestamp. An alert message indicating “Tamper Alert – Device Removed” is then transmitted immediately to the designated emergency contact through the Telegram platform [2]. Even if the device is subsequently destroyed, disconnected, or powered off, the captured image and location data are already transmitted and stored on the remote Telegram server, ensuring that critical evidence remains preserved for further investigation.

### *B. Autonomous Evidence Generation*

Each emergency alert generated by the system contains multiple pieces of contextual information to provide comprehensive situational awareness [4], [8]. The transmitted alert includes the following components:

- Emergency type (SOS, Loud Sound, or Tamper Alert)
- Real-time image evidence captured by the ESP32-CAM [1]
- Timestamp indicating the exact date and time of the event
- GPS location link in Google Maps format [15]

By combining these elements, the system generates multimodal evidence that can assist emergency responders in quickly identifying the situation and the location of the wearer.

### *C. Example Alert Format*

The following example illustrates the structure of an alert message received through the Telegram platform [2]:

```
Emergency Alert
Loud Sound Distress
28-11-2025, 22:41:08
https://maps.google.com/?q=12.9732,77.5914
Image: [Attached]
```

## VI. Results and Discussion

The prototype of the proposed wearable safety device was implemented and experimentally evaluated under multiple emergency scenarios corresponding to each trigger type. The testing procedure aimed to validate the functional performance of the system and assess its reliability in real-world conditions. The evaluation focused on key performance aspects including alert transmission time, GPS location accuracy, sensor reliability, and the integrity of transmitted evidence.

### *A. Alert Transmission Time*

The response time of the system was measured as the delay between trigger activation and the reception of the alert message on the Telegram platform [2]. Experimental observations showed that the alert transmission typically occurred within a few seconds, depending on network conditions and the size of the captured image. This delay includes several sequential operations such as trigger detection, image capture by the ESP32-CAM [1], acquisition of GPS coordinates, message formatting, and data transmission through the Telegram

Bot API [2]. Despite these processing steps, the total delay remained sufficiently low for practical emergency response scenarios, consistent with findings in related IoT-based safety research [3], [6]. The rapid delivery of alerts ensures that emergency contacts receive situational information almost immediately after an incident occurs.

### B. GPS Accuracy

The GPS module used in the system provided location coordinates with an accuracy of a few meters under open outdoor conditions [15]. During testing in semi-urban environments with moderate signal obstruction, the module continued to produce reliable location estimates that were adequate for identifying the approximate location of the wearer. To simplify interpretation by the recipient, the coordinates were automatically converted into a clickable Google Maps link included within the alert message. This feature enables emergency contacts to quickly visualize the location of the incident without requiring additional processing or manual coordinate entry.

### C. Sensor Calibration

The performance of the sensing components was evaluated through calibration and controlled testing. The microphone sensor threshold was adjusted to detect high-intensity distress sounds such as shouting or calls for help while minimizing false alarms caused by normal environmental noise.

Similarly, the SW-420 vibration sensor was calibrated to respond only to strong and sustained vibrations that may occur during forced removal or tampering with the device. Minor movements and routine user activities did not trigger alerts. During testing, tamper alerts were successfully generated whenever the device was forcefully pulled, shaken, or detached, demonstrating the reliability of the sensor configuration.

### D. Evidence Reliability

A key advantage of the proposed system is the secure preservation of incident evidence. Once an emergency trigger is activated, the captured image, timestamp, and GPS location data are immediately transmitted to the Telegram server through the cloud-based messaging platform. This ensures that the evidence is stored remotely and remains accessible even if the wearable device is damaged, disconnected, or destroyed after the event.

The ability to store visual and contextual data on a remote server enhances the reliability of the system and supports post-incident investigation by providing verifiable records of the event.

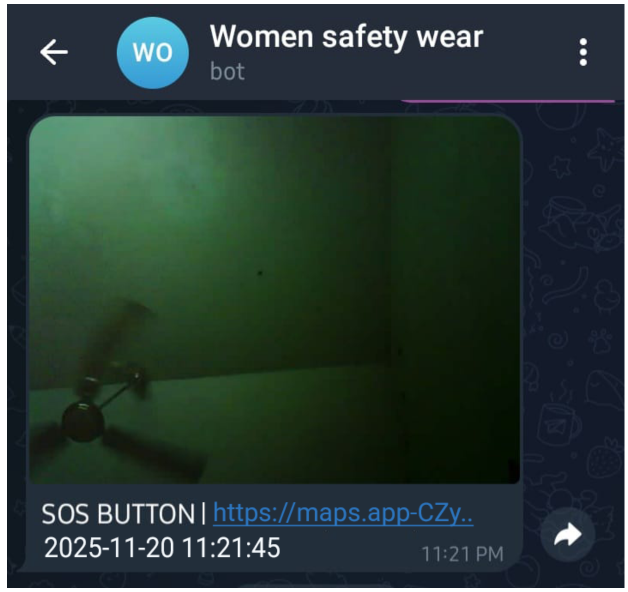


Fig. 3. Telegram SOS notification generated by the wearable device.

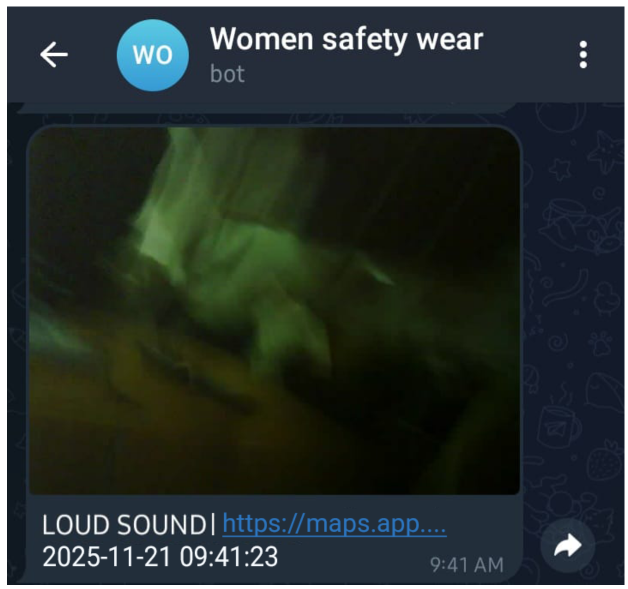


Fig. 4. Telegram message triggered by loud-sound detection.

## VII. Future Scope

he present implementation focuses on three primary triggers and basic evidence capture. Several enhancements can be incorporated in future work:

- Integration of TinyML or edge AI models could allow detection of specific keywords, screams, or suspicious patterns in audio and video data, thereby improving the intelligence of emergency detection [4], [14].
- Adding GSM or LTE modules as a backup communication channel would enable alert transmission even in the absence of Wi-Fi [15].
- Storing alerts in a dedicated cloud database can facilitate

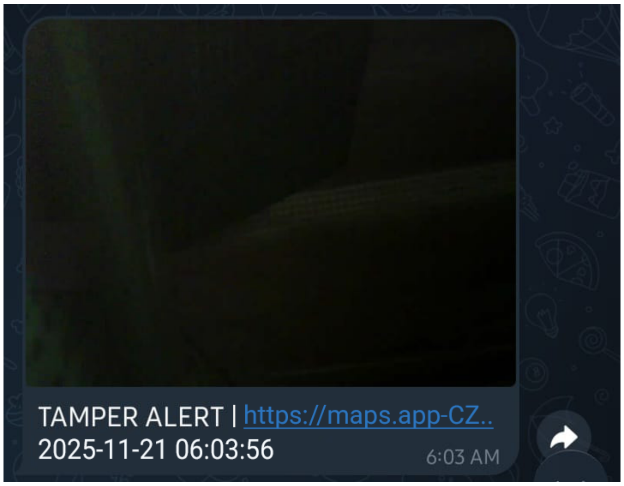


Fig. 5. Telegram message triggered by device tampering.

long-term logging, analytics, and retrieval for legal or investigative purposes [8].

- A feedback channel from the responder to the device could acknowledge alert reception and possibly provide audio guidance to the victim [13].
- The system could be extended to interface with police portals or national emergency numbers, enabling automated escalation in critical situations [11], [16].

## VIII. Conclusion

This paper presents a wearable emergency detection system that employs an event-driven architecture to provide autonomous, multi-modal evidence generation and rapid alert transmission. By combining three distinct triggers—SOS button activation, loud-sound distress detection, and tamper or forced removal detection—the device ensures that the majority of practical emergency scenarios are covered [5]–[7]. The integration of ESP32-CAM [1] for real-time image capture, GPS modules for location tracking [15], and RTC for precise time stamping results in rich, contextual data that can significantly aid in timely intervention and post-incident investigation. The use of the Telegram Bot API [2] for communication makes the system globally accessible and leverages existing cloud infrastructure for secure evidence storage.

Experimental results demonstrate acceptable latency, robust tamper detection, and satisfactory GPS accuracy for real-world use [3], [4]. Owing to its low cost, portability, and scalability, the proposed system is suitable for deployment among students, women, elderly individuals, night-shift workers, and other vulnerable groups [8], [11], [12]. Compared with conventional safety applications and standalone GPS tracking devices, the proposed system offers enhanced situational awareness through multi-modal evidence generation [13], [16]. The integration of automated sensing, visual documentation, and cloud-based alert transmission represents a significant step toward intelligent wearable safety technologies..

## Acknowledgment

The authors would like to thank the supporting staff, peers, and institution for their guidance and encouragement during the development and testing of this project.

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